Agenda

Propagation error: short review

Linear combinations of measurements

Point estimation

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Summary

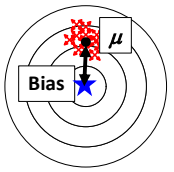
- A measured value is a random variable with mean μ and standard deviation σ.
- The bias in the measuring process is the difference between the mean measurement and the true value:

$$\text{Bias} = \mu - \text{true value}$$

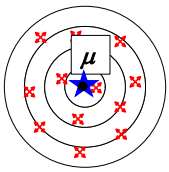
- The uncertainty in the measuring process is the standard deviation σ.
- The smaller the bias, the more accurate the measuring process.
- The smaller the uncertainty, the more precise the measuring process.

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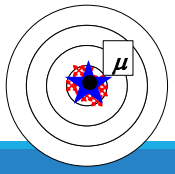
Propagation of error



Low uncertainty
Non-zero bias



High uncertainty
No bias



Low uncertainty
No bias

Target → True value ★

Shots → Measurements ✕

Average of shots → Mean measurement (μ) •

Spread of shots → Uncertainty (σ)

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Concepts to be repeated in different contexts

We will see the same concepts in point estimation as well.

- **Systematic error (bias) - accuracy**
- **Random error - precision**

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Linear Combinations of Measurements

X: a random variable, $\mu = E(X)$ and $\sigma^2 = \mathrm{Var}(X)$

$Y = a + bX$: function of X

$E(Y)$ and $\mathrm{Var}(Y)$: ?

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Linear Combinations of Measurements

$(X_1, X_2, \cdots, X_n)$: a random sample with common mean μ and variance σ^2

$Y = \sum_{i=1}^{n} c_i X_i$: function of $(X_1, X_2, \cdots, X_n)$, where c_i's are constants

$E(Y)$ and $\mathrm{Var}(Y)$: ?

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Example

A surveyor is measuring the perimeter of a rectangular lot. He measures two adjacent sides to be 50.11 ± 0.05m and 75.12 ± 0.08m. These measurements are independent. Estimate the perimeter of the lot and find the uncertainty in the estimate.

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Linear Combinations of Measurements

$(X_1, X_2, \cdots, X_n)$: a random sample with common mean μ and variance σ^2

Sample mean:

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

$E(\bar{X})$ and $\mathrm{Var}(\bar{X})$: ?

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Example

The length of a component is to be measured by a process whose uncertainty is 0.05 cm. If 25 independent measurements are made and the average of these is used to estimate the length, what will the uncertainty be? How much more precise is the average of 25 measurements than a single measurement?

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Linear Combinations of Measurements

X: a measurement with a certain device with mean μ_X and variance σ_X^2

Y: another (independent) measurement with another device with mean μ_Y and variance σ_Y^2

$Z = cX + (1 - c)Y$

Can we find the value of c that minimizes $\mathrm{Var}(Z)$?

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Linear Combinations of Measurements

(X_1, X_2): a possibly correlated sample with mean μ_i and variance σ_i^2

$Y = c_1X_1 + c_2X_2$: function of (X_1, X_2), where c_i's are constants

$E(Y)$ and $\mathrm{Var}(Y)$: ?

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Linear Combinations of Measurements

$(X_1, X_2, \cdots, X_n)$: a possibly correlated sample with mean μ_i and variance σ_i^2

$Y = \sum_{i=1}^{n} c_i X_i$: function of $(X_1, X_2, \cdots, X_n)$, where c_i's are constants

$E(Y)$ and $\mathrm{Var}(Y)$: ?

$$E(Y) = E(\sum_{i=1}^{n} c_i X_i) = \sum_{i=1}^{n} c_i \mu_i$$

$$Var(Y) = Var(\sum_{i=1}^{n} c_i X_i) = \sum_{i=1}^{n} c_i^2 \sigma_i^2 + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} c_i c_j Cov(X_i, X_j)$$

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Linear Combinations of Measurements

$(X_1, X_2, \cdots, X_n)$: a possible correlated sample with mean μ_i and variance σ_i^2

$Y = \sum_{i=1}^{n} c_i X_i$: function of $(X_1, X_2, \cdots, X_n)$, where c_i's are constants

An upper bound on σ_Y : ?

$$\sigma_Y \leq \sum_{i=1}^{n} |c_i| \sigma_i$$

See your book for the details of the derivation.

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Example

A surveyor is measuring the perimeter of a rectangular lot. He measures two adjacent sides to be 50.11 ± 0.05 m and 75.21 ± 0.08 m. These measurements are not necessarily independent. Find a conservative estimate of the uncertainty in the perimeter of the lot.

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Linear Combinations of Measurements

Two independent random samples are observed:

$(X_1, X_2, \cdots, X_n)$: a random sample with common mean μ_X and variance σ_X^2

$(Y_1, Y_2, \cdots, Y_m)$: a random sample with common mean μ_Y and variance σ_Y^2

$Z = a\bar{X} + b\bar{Y}$

$E(Z)$ and $\mathrm{Var}(Z)$: ?

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Nonlinear Functions of Measurements

See your book if interested. We skip this section in this course.

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Summary

We discussed measurement error.

Main concepts:

- Different contributions to measurement error: bias and uncertainty
- Linear combinations of measurements:
  - Independent measurements with equal and different uncertainties
  - Dependent measurements

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Class exercise

Consider the random variables X and Y with the following joint probability mass function:

$P_{X,Y}(1,1) = P_{X,Y}(0,0) = \frac{3}{8}$; $P_{X,Y}(0,1) = P_{X,Y}(1,0) = \frac{1}{8}$

a) (E[X], E[Y])=?

b) (Var(X), Var(Y))=?

c) Cov(X,Y)=?

d) Var(2X+3Y)=?

$Var(X) = E[(X - \mu_X)^2] = E[X^2] - (\mu_X)^2$
$Cov(X,Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X \mu_Y$
$Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2abCov(X,Y)$

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